

VNUHCM - University of Science
Faculty of Information Technology

CS486 – Lecture 3

Relational Data Model

Le Thi Nhan
ltanh@fit.hcmus.edu.vn

Content

- Introduction
- Concept
- Constraint
- From E/R diagram to relational design

Introduction

- Was first introduced by E. F. Codd
 - “A Relation Model for Large Shared Data Banks”, Communications of ACM, 1970

- Commercial implementation
 - By IBM
 - System R (1974), SQL/DS (1981), DB2 (1983)
 - Oracle (1979)
 - By Sybase
 - SQL Server (1987), Adaptive Server Enterprise (1996)
 - By Microsoft
 - SQL Server (1989)
 - Access (1992)

Introduction

- Open source implementation
 - MySQL
 - By MySQL AB, 1995
 - PostgreSQL
 - Ingres project at the University of California, Berkeley, 1980s
 - By many developers, released in 1996
 - SQLite
 - By D. Richard Hipp working for General Dynamics, 2000

Introduction

- Provide a simple way to represent data
 - The relation: a two-dimensional table
- The theoretical background
 - Set theory of mathematical logic

Content

- Introduction
- **Concept**
 - Relation
 - Attribute
 - Schema
 - Tuple
 - Domain
 - Characteristics of relation
 - Notations
- Constraint
- From ER diagram to relational design

Relation

- Relational model presents the DB as a collection of ***relations***
 - A relation = a two-dimensional table

Each column is one of the attributes of the entity set

FName	LName	BirthDate	Address	Sex	Salary	DNo
John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
Joyce	English	07/31/1972	5631 Rice, Houston, TX	Female	25000	5

Each row is one employee entity

Relation name is EMPLOYEE

Relation

- Includes
 - Name
 - Set of columns
 - Fixed
 - Named
 - Has data types
 - Set of rows
 - Changed by time
- A row ~ A real-world entity or relationship
- A relation ~ An entity set or relationship

Attribute

- The names for columns of the relation
- Describes the meaning of entries in the column below

Attributes

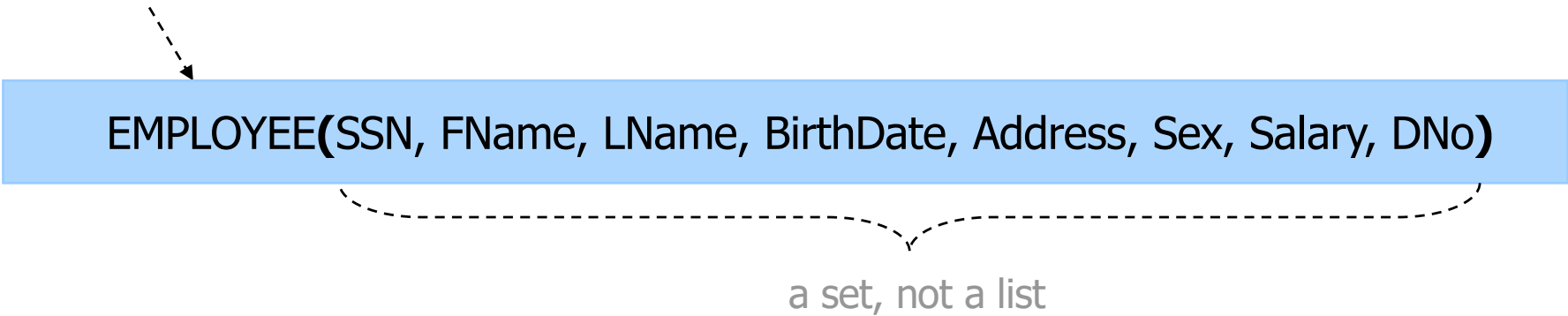
FName	LName	BirthDate	Address	Sex	Salary	DNo
John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
Joyce	English	07/31/1972	5631 Rice, Houston, TX	Female	25000	5

- All values in a column are of the same data type

Schema

- Schema of a relation
 - Name
 - Set of attributes

Relation schema



EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary, DNo)

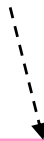
The diagram illustrates a relation schema. A dashed arrow points from the text 'Relation schema' to a light blue rectangular box containing the text 'EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary, DNo)'. Below this box, a dashed curly brace spans the width of the attribute list, with the text 'a set, not a list' centered underneath it.

a set, not a list

Schema

- Database schema
 - A design consist of one or more relational schemas

Database schema



```
EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary, DNo)
DEPARTMENT(DNumber, DName, MgrSSN, MgrStartDate)
DEPT_LOCATION(DNumber, DLocation)
DEPENDENT(SSN, Dependent_Name, Sex, BDate, Relationship)
PROJECT(PName, PNumber, PLocation, DNum)
```

Tuple

- Row of a relation
 - Except the header row containing the attribute names
- Contains many components
 - One component for each attributes of the relation

<John, Smith, 01/09/1965, "731 Fondren, Houston, TX" , Male, 30000, 5>



a component

Domain

- Each attribute of a relation associates with a ***domain***
 - A particular elementary type
- A component of each tuple
 - Is ***atomic***
 - Has a ***value*** that belongs to the domain of the corresponding attribute
- Example
 - FName: string, DOM(FName): the set of strings
 - Salary: integer, DOM(Salary): the set of integers

Characteristics of relation

- The order of tuples in a relation is not important

FName	LName	BirthDate	Address	Sex	Salary	DNo
John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
Joyce	English	07/31/1972	5631 Rice, Houston, TX	Female	25000	5

- The order of values in a tuple is important

<John, Smith, 01/09/1965, "731 Fondren, Houston, TX" , **Male, 30000**, 5>

Differs from

< John, Smith, 01/09/1965, "731 Fondren, Houston, TX", **30000, Male**, 5>

Characteristics of relation

- Each value of components in a tuple
 - Atomic or
 - NULL
- Relations are sets of tuples, not lists of tuples
 - There are no identical tuples

Relational model notation

■ Relation schema

- Given $A_1, A_2, \dots, A_n$ are attributes
- Has domains $D_1, D_2, \dots, D_n$ respectively
- Is denoted by $R(A_1:D_1, A_2:D_2, \dots, A_n:D_n)$
- Example
 - EMPLOYEE(SSN:DOM(integer), FName:DOM(string), LName:DOM(string), Birthday:DOM(date), Address:DOM(string), Sex:DOM(string), Salary:DOM(integer), DNo:DOM(integer))

■ The **degree** of a relation is the number of attributes of its relation schema

- EMPLOYEE is a relation schema of degree 8

Relational model notation

■ Relation instances

- A relation instance r of a relation schema $R(A_1, A_2, \dots, A_n)$, denoted by $r(R)$, is a set of tuples $r = \{t_1, t_2, \dots, t_k\}$
- Where each t_i is an ordered list of n values $t_i = \langle v_1, v_2, \dots, v_n \rangle$
 - Each v_j is a member of $DOM(A_j)$ or NULL value

	FName	LName	BirthDate	Address	Sex	Salary	DNo
t_1	John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
t_2	Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
t_3	Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
t_4	Joyce	English	07/31/1972	NULL	Female	25000	5

v_i

Summary of denotations

- The relation schema R of the degree n
 - $R(A_1, A_2, \dots, A_n)$
- The attribute set of R
 - R^+
- Relations
 - R, S, P, Q
- Tuples
 - t, u, v
- The domain of the attribute A
 - $DOM(A)$
- The value at the attribute A of the t^{th} tuple
 - $t.A$ or $t[A]$

Content

- Introduction
- Concept
- **Constraint**
 - Superkey
 - Key
 - Primary key
 - Reference
 - Foreign key
- From ER diagram to relational design

Constraint

- Integrity constraint
 - Rules, conditions need to satisfy for all of instances of relational database
- Constraints
 - Defined when the relation schema is modeled
 - Checked when the data in relations are modified

Superkey

■ Definition

- Assume SK is a subset of attributes of R , $SK \neq \emptyset$
- SK is the super key if

$$\forall r, \forall t1, t2 \in r, t1 \neq t2 \Rightarrow t1[SK] \neq t2[SK]$$

Any two distinct tuples have the different values at the superkey

■ Remark

- No two tuples in any state r of R can have the same value for superkey
- Every relation has at least one default superkey

Example

- Find all superkeys of R

R	A	B	C	D
	x	1	10	a
	x	2	20	a
	y	1	40	b
	y	1	40	c
	z	1	50	d

Key

■ Definition

- Assume K is a subset of attributes of R , $K \neq \emptyset$
- K is a key if
 - **K is a superkey of R and**
 - **$\forall K' \subset K, K' \neq K, K'$ is not the superkey of R**

A key is the minimal superkey

■ Remark

- The value of a key identifies uniquely each tuple in the relation
- A key is a *property* of the relation schema
 - Time-invariant: a constraint should hold on every valid state
- A key is determined from the meaning of attributes
- A relation has more than one key

Primary key (PK)

- Designate one of the keys as the primary key
 - The value for *PK* is constrained to be not null
 - Underline the attributes of *PK* when displaying its relation schema
- The choice of *PK*
 - Influence some implementation issues
 - Usually with a single attribute or a small number of attributes

Reference

- R refers to S when
 - An attribute A of a tuple in relation R receives a value from an attribute B of relation S
 - Must refer to an existing tuple

		S				
		DName	<u>DNumber</u>			
		Research	5			
		Headquarters	4			
		Administration	1			

R	FName	LName	BirthDate	Address	Sex	Salary	DNo
	John	Smith	01/09/1965	731 Fondren, Houston, TX	Male	30000	5
	Alicia	Zelaya	12/08/1955	3321 Castle, Spring, TX	Female	25000	4
	Jennifer	Wallace	06/20/1941	291 Berry, Bellaire, TX	Female	43000	4
	Joyce	English	07/31/1972	5631 Rice, Houston, TX	Female	25000	5

Foreign key (FK)

- Examine two relation schemas R and S
 - Assume FK is a set of attributes of R , $FK \neq \emptyset$
 - FK is a foreign key of R if
 - Attributes in FK have the same domains as the primary key attributes PK of S
 - A value of FK in a tuple $t_1 \in R$
 - * Either is a value of PK for some tuple $t_2 \in S$
 - * Or is null

■ Example

EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary, DNo)

DEPARTMENT(DName, DNumber)

Primary key

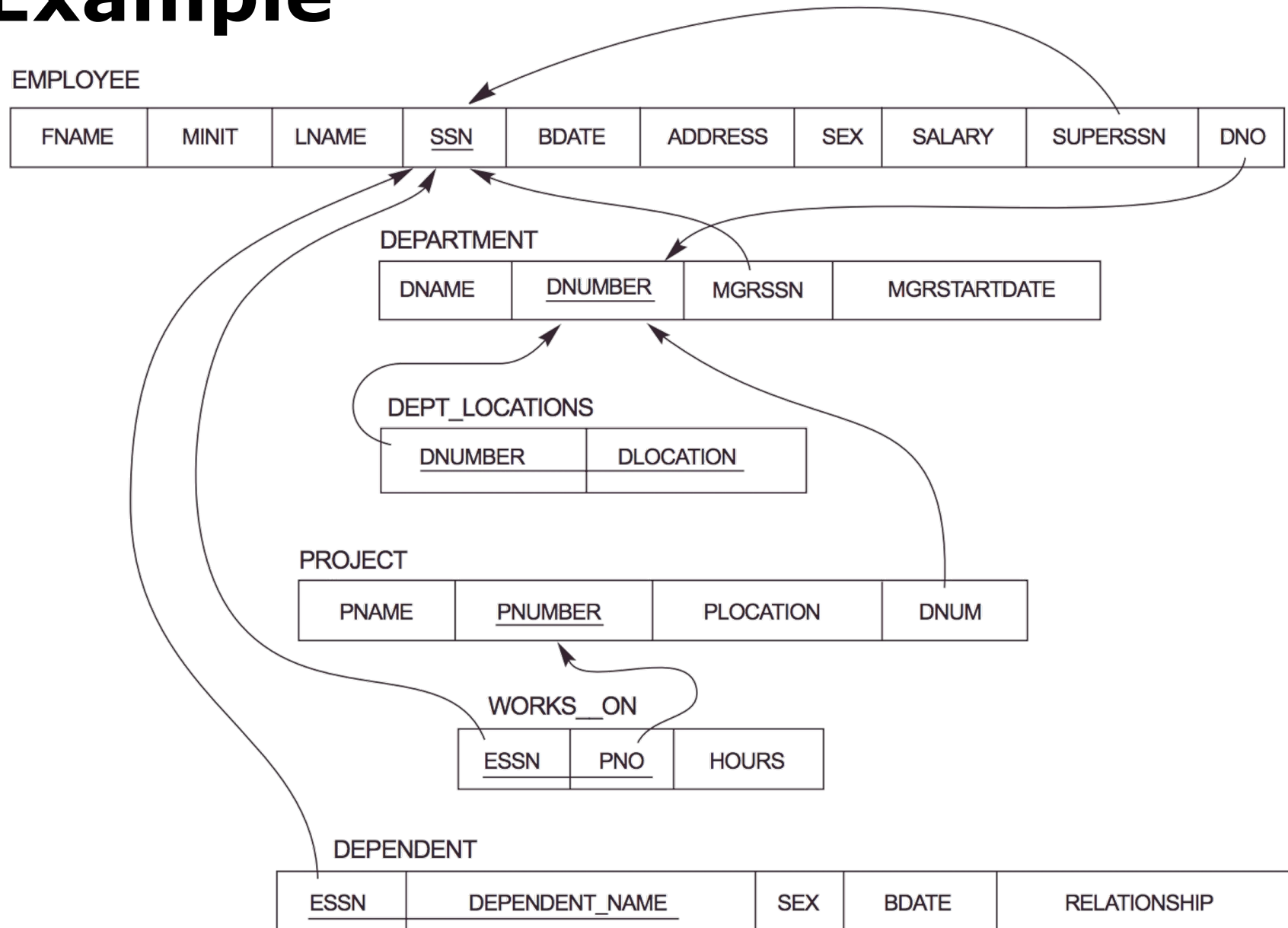
Foreign key

Foreign key

■ Remark

- An attribute can both participate in *PK* and participate in *FK*
- A *FK* can refer to its own relation
- Many *FKs* might refer to the same primary key
- Referential constraint = Foreign key constraint

Example



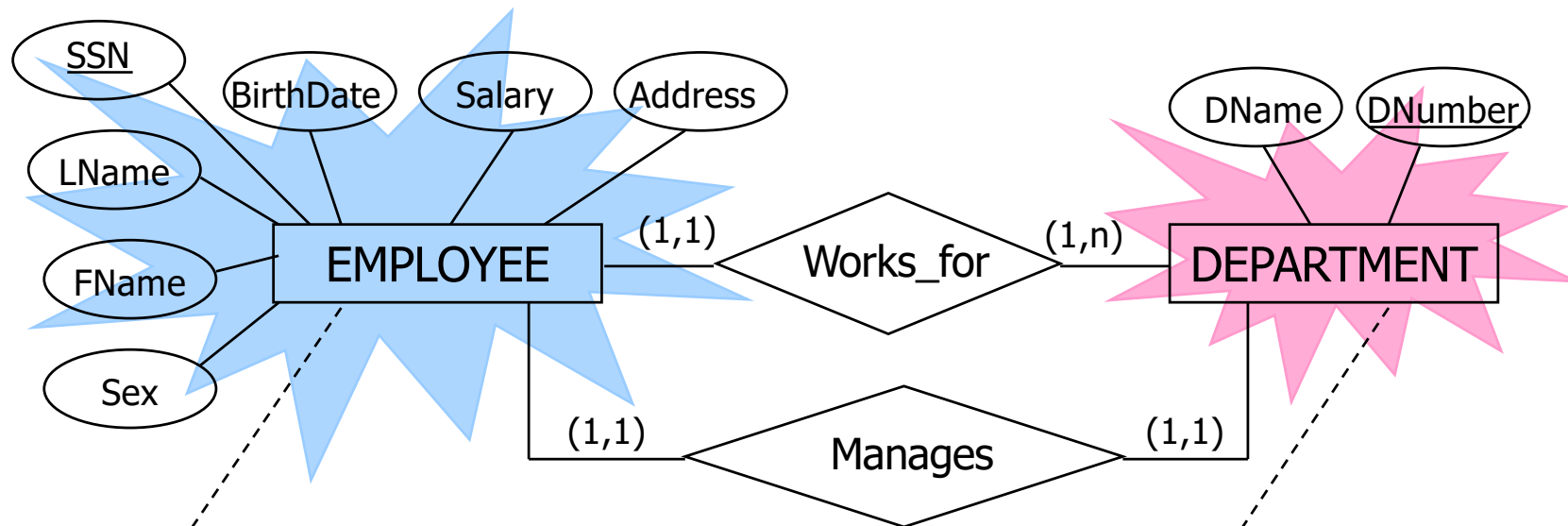
Content

- Introduction
- Concepts
- Constraints
- **From E/R diagrams to relational design**
 - Rules

Rules

■ (1) Entity set

- Turn each entity set (except weak entity set) into a relation with the same set of attributes



EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary)

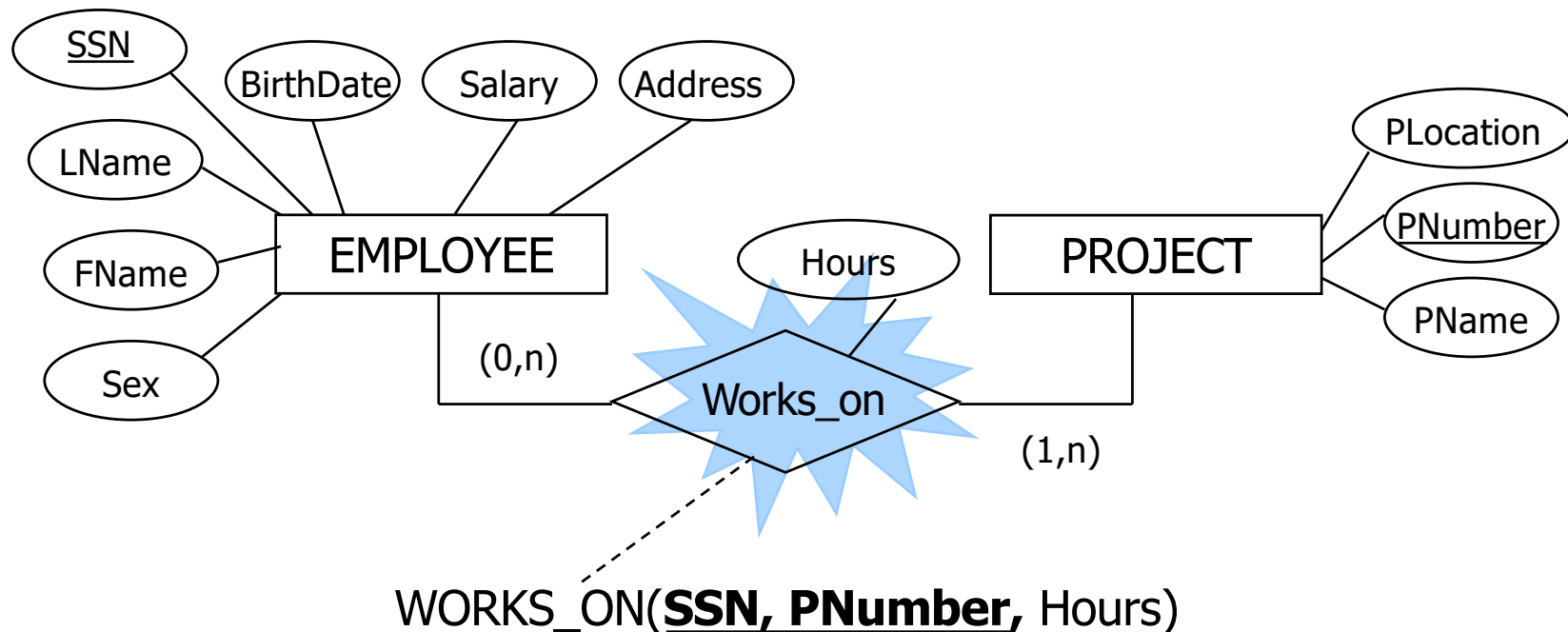
DEPARTMENT(DName, DNumber)

Rules

■ (2) Relationship

- (2a) Many-Many

- Create a new relation
 - * Relation name is the name of the relationship
 - * Attributes are the key attributes of connected entity sets

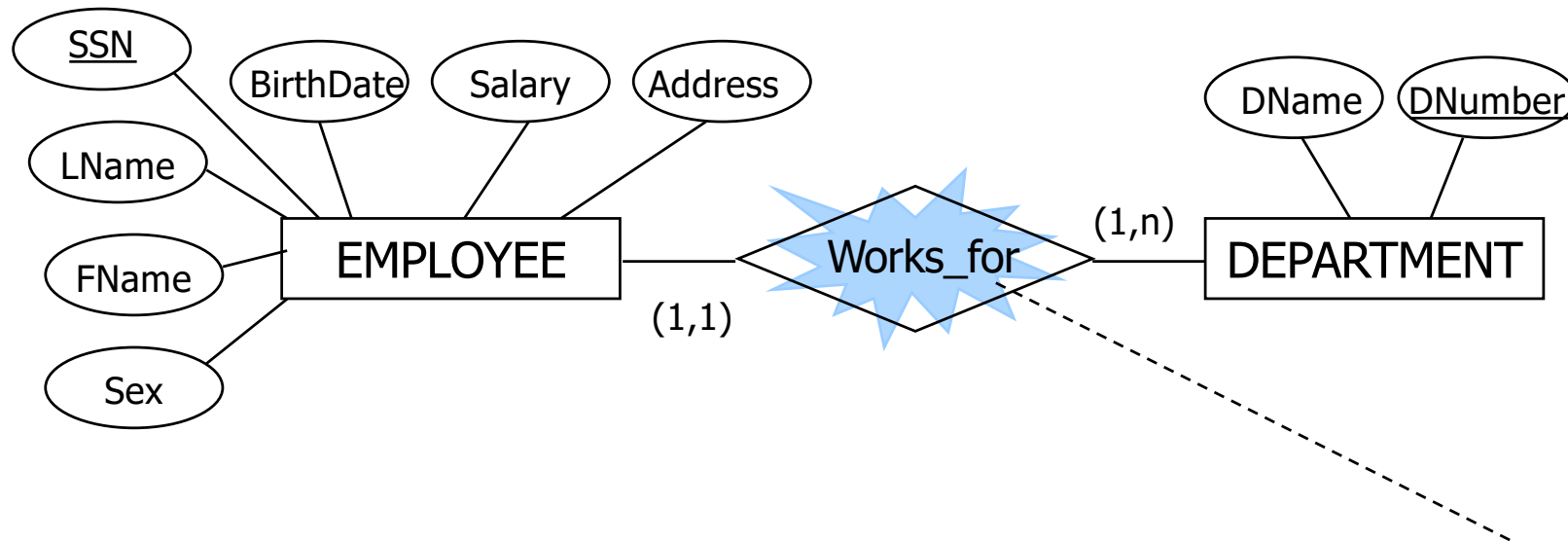


Rules

■ (2) Relation

- (2b) One-Many

- Adding the key of the many-relation to the one-relation



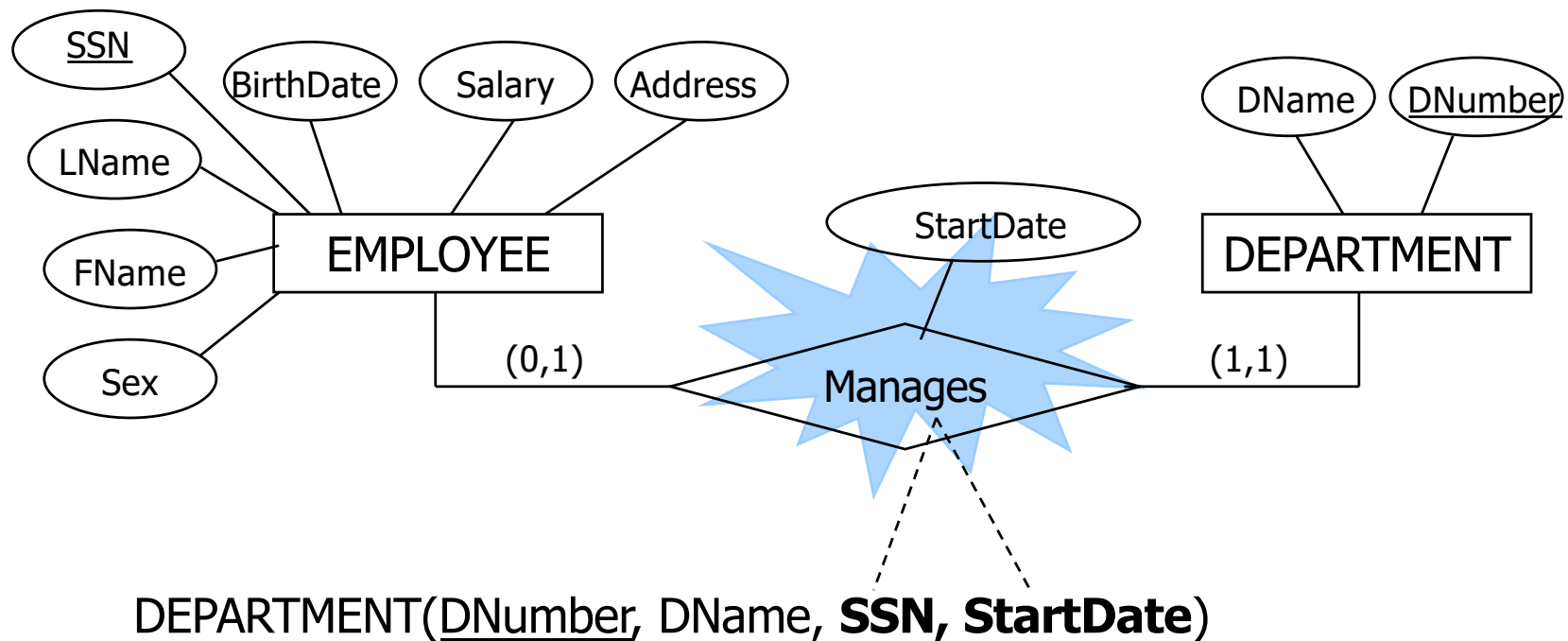
EMPLOYEE(SSN, FName, LName, BirthDate, Address, Sex, Salary, **Dnumber**)

Rules

■ (2) Relationship

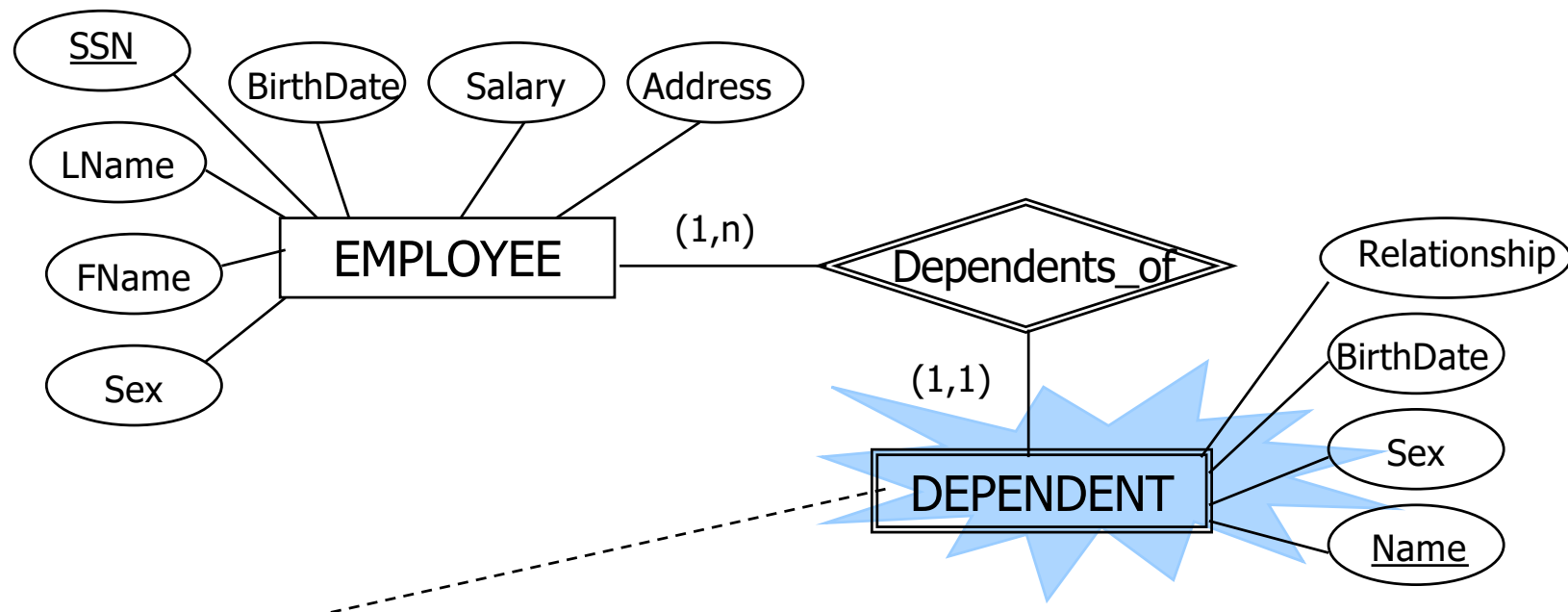
- (2c) One-One

- Either adding the key of a relation to another relation
- Or adding the key to both relations



Rules

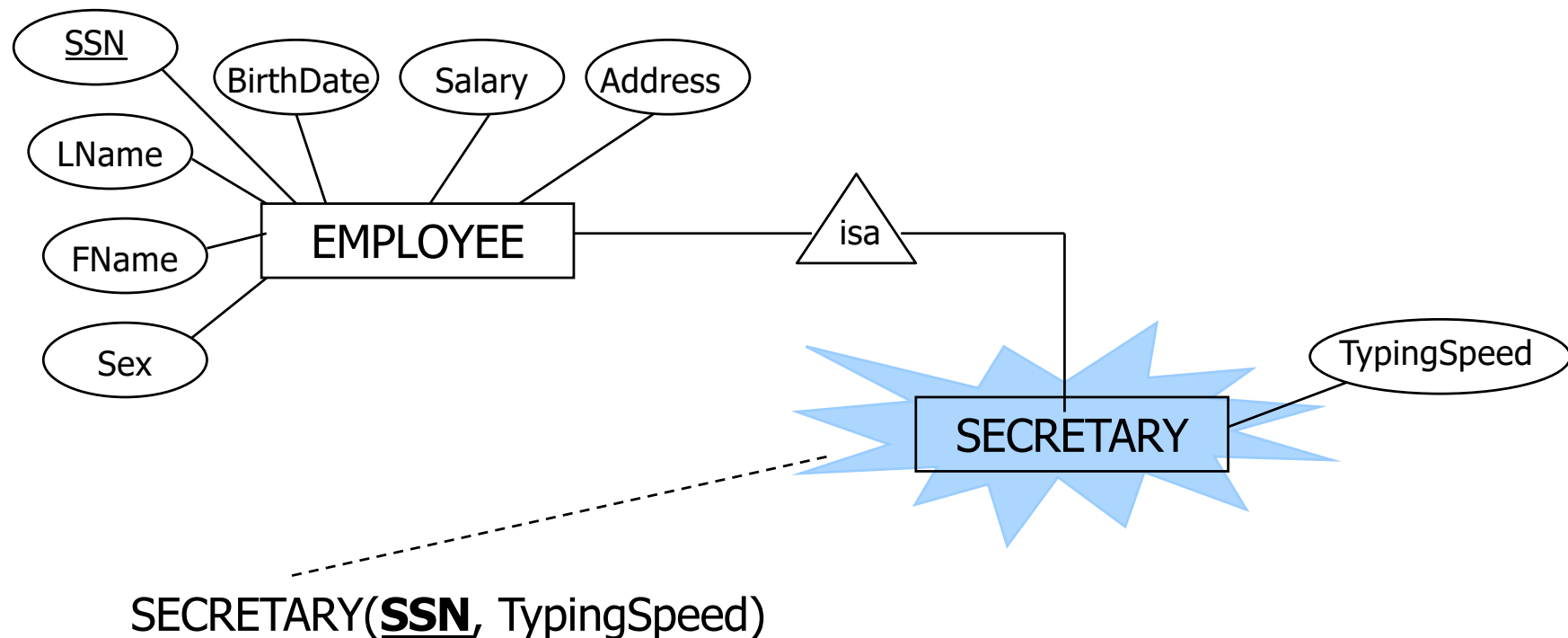
- (3) Weak entity set
 - Turn into a relation
 - Has the same name
 - Add the key of related entity sets



DEPENDENT(SSN, Name, Sex, BirthDate, Relationship)

Rules

- (4) Subclass
 - Turn into a relation
 - Has the same name
 - Add the key of the superclass



Next lecture

- Query languages
- Read
 - Chapter 2, 5, 6, 8 [1]
 - Chapter 6 [2]
- Watch
 - Video clip

